

1. Introduction

Nuclear power plants generate electricity by converting thermal energy from a nuclear heat source into electrical energy. Although the nuclear reaction itself is outside the scope of this project, the subsequent conversion of heat into electricity involves several areas of chemical and process engineering, including thermodynamics, heat transfer, fluid flow, process control and process safety.

This project develops a simplified process-engineering assessment of a hypothetical 1 GW nuclear power plant. The assessment uses an Excel-based model to investigate the relationship between electrical output, thermal efficiency, waste-heat rejection and annual electricity generation. A comparison of indicative lifecycle greenhouse-gas emissions with natural gas generation is also included.

The project is intended as an engineering assessment rather than a detailed nuclear plant design. Simplifications and assumed values are therefore used to demonstrate the relevant engineering principles while keeping the model suitable for an initial feasibility-level analysis.

2. Project Objective

The main objective of the project was to develop a simplified energy model for a hypothetical 1 GW nuclear power plant and use it to investigate key process-engineering relationships.

The specific objectives were to:

- calculate the thermal power required to produce a specified electrical output at a given thermal efficiency;
- determine the resulting waste-heat load;
- estimate annual electricity generation using an assumed plant capacity factor;
- investigate the effect of thermal efficiency and capacity factor through sensitivity analysis;
- compare indicative annual lifecycle emissions with natural gas generation; and
- identify key chemical engineering considerations associated with nuclear power generation, including heat transfer, fluid flow, process control, process safety and radioactive waste management.

3. Process Description

The simplified process considered in this assessment represents the conversion of heat from a nuclear source into electrical power through a steam cycle. The nuclear heat source

provides thermal energy to a steam-generation system, where water is heated to produce steam. The steam is then expanded through a turbine, converting thermal energy into mechanical work.

The turbine drives an electrical generator, which converts the mechanical energy into electrical energy. For the hypothetical plant considered in this project, the target electrical output is 1 GW.

After expanding through the turbine, the steam is directed to a condenser, where it rejects heat and is converted back into liquid water. A feedwater pump then increases the pressure of the water before it is returned to the steam-generation stage. This creates a continuous water and steam cycle.

The simplified process can therefore be represented as:

Nuclear heat source → Steam generation → Steam turbine → Electrical generator → Electricity

with the supporting cycle:

Steam turbine → Condenser → Feedwater pump → Steam generation

The condenser represents an important source of waste-heat rejection. Consequently, the thermal efficiency of the overall power-generation process has a direct effect on the amount of heat that must ultimately be rejected to the environment.

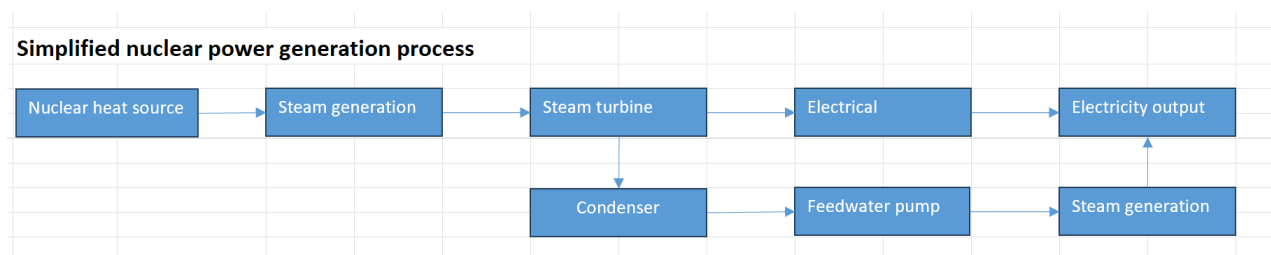


Figure 1: Simplified nuclear power generation process considered in the assessment.

4. Engineering Method

The assessment was developed using an Excel-based calculation model. The model uses a hypothetical electrical output of 1,000 MW, a thermal efficiency of 33% and a capacity factor of 91% as the base-case assumptions.

4.1 Thermal Power Requirement

The thermal power required to produce the target electrical output was calculated using the relationship:

$$\eta = P_e / P_t$$

where η is the thermal efficiency, P_e is the electrical power output and P_t is the required thermal power input.

Rearranging gives:

$$P_t = P_e / \eta$$

Using an electrical output of 1,000 MW and a thermal efficiency of 33%:

$$P_t = 1000 / 0.33 = 3030.3 \text{ MW}$$

Therefore, approximately 3,030 MW of thermal power is required to produce the specified 1,000 MW electrical output under the assumptions of the model.

4.2 Waste Heat

The difference between the thermal power supplied and the electrical power produced was used as a simplified estimate of the heat that must be rejected:

$$Q_{\text{waste}} = P_t - P_e$$

Therefore:

$$Q_{\text{waste}} = 3030.3 - 1000 = 2030.3 \text{ MW}$$

This provides a simplified estimate of the thermal energy that must ultimately be rejected or is otherwise lost from the overall energy-conversion process. It therefore highlights the importance of heat rejection and cooling systems in thermal power generation

4.3 Annual Electricity Generation

Annual electricity generation was estimated using the plant electrical capacity, the number of hours in a year and the assumed capacity factor:

$$E = P_e \times H \times CF$$

where E is annual electricity generation, H is the number of hours per year and CF is the capacity factor.

Using 8,760 hours per year and a capacity factor of 91%:

$$E = 1000 \times 8760 \times 0.91$$

This gives an annual generation of approximately **7.97 TWh/year**.

4.4 Indicative Emissions Comparison

An indicative emissions comparison was also performed using assumed lifecycle emissions intensities for nuclear and natural gas electricity generation.

Annual emissions were calculated by multiplying annual electricity generation by the relevant emissions intensity. The annual generation value was converted to kWh before applying the emissions intensity.

The model used an assumed nuclear emissions intensity of **12 g CO₂e/kWh** and a natural gas intensity of **400 g CO₂e/kWh**. The resulting emissions were converted from grams to tonnes.

For the base case, the model estimates approximately **95,659 tonnes CO₂e/year** for nuclear generation and **3,188,640 tonnes CO₂e/year** for an equivalent amount of natural gas generation.

These values are indicative rather than plant-specific measurements and depend directly on the assumed emissions intensities.

5. Results

The base-case model gives a required thermal input of approximately 3,030 MW for a 1,000 MW electrical output, corresponding to an estimated waste-heat load of 2,030 MW. At a capacity factor of 91%, annual electricity generation is approximately 7.97 TWh.

Using the assumed lifecycle emissions intensities, the model estimates approximately 95,659 tonnes CO₂e/year for nuclear generation compared with approximately 3.19 million tonnes CO₂e/year for the equivalent natural gas generation.

The main results are summarised in the Excel workbook, alongside the sensitivity analysis and process diagram.

6. Sensitivity Analysis

Sensitivity analysis was carried out to investigate how changes in thermal efficiency and capacity factor affect the performance of the hypothetical plant.

6.1 Effect of Thermal Efficiency

Thermal efficiency was varied between 25% and 40%, while the electrical output was maintained at 1,000 MW. The results show that increasing thermal efficiency reduces both the required thermal power input and the resulting waste-heat load.

At an efficiency of 25%, approximately 4,000 MW of thermal power would be required, resulting in approximately 3,000 MW of waste heat. At 40% efficiency, the required thermal power falls to 2,500 MW and the estimated waste heat falls to 1,500 MW.

Figure 2 shows the effect of thermal efficiency on the estimated waste-heat load. The trend demonstrates the importance of thermal efficiency when considering the heat-rejection requirements of a large thermal power plant.

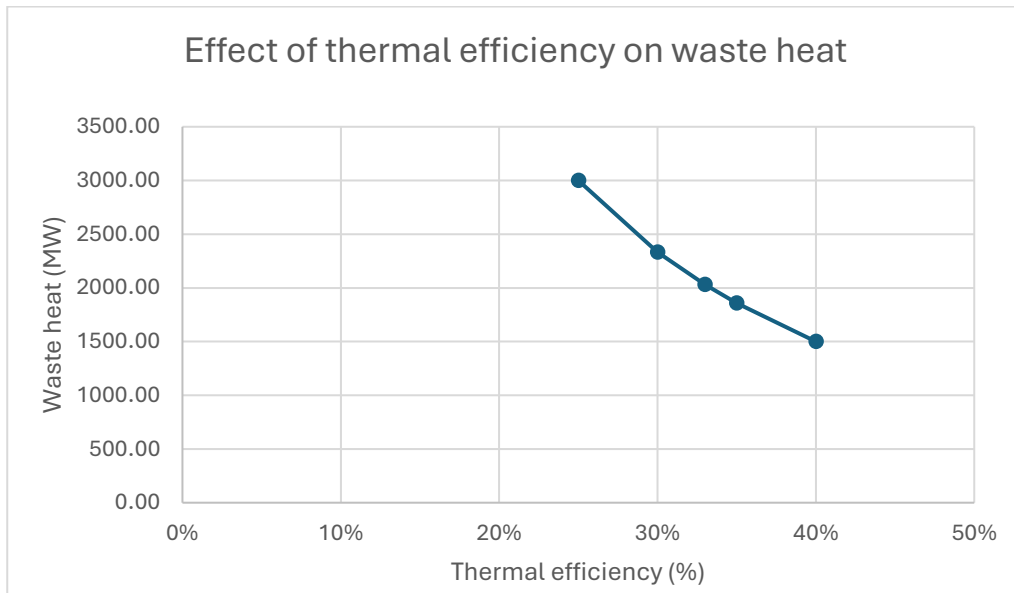


Figure 2

6.2 Effect of Capacity Factor

Capacity factor was varied between 70% and 95% while maintaining the plant's rated electrical output at 1,000 MW. Unlike thermal efficiency, changing the capacity factor does not change the plant's rated electrical capacity. Instead, it changes the amount of electricity generated over a year.

The model estimates annual generation of approximately 6.13 TWh at a 70% capacity factor, increasing to approximately 8.32 TWh at 95%. The base-case value of 91% gives approximately 7.97 TWh/year.

Figure 3 shows the approximately linear relationship between capacity factor and annual electricity generation. The sensitivity analysis therefore demonstrates how operational availability can influence annual energy production even when the rated capacity of the plant remains unchanged.

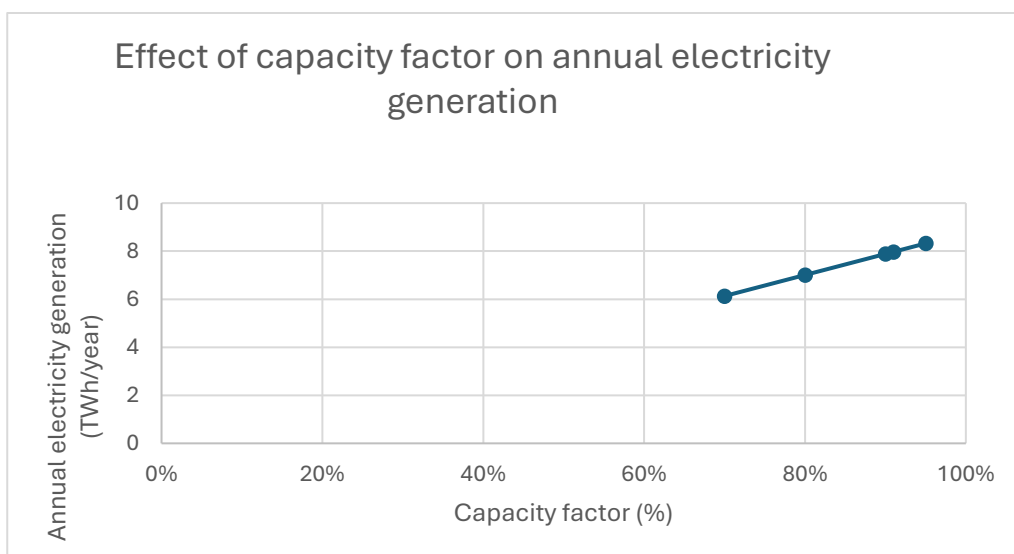


Figure 3

7. Chemical Engineering Considerations

Although nuclear power generation involves nuclear engineering alongside several other engineering disciplines, the conversion of nuclear heat into electricity contains a number of processes relevant to chemical engineering.

7.1 Thermodynamics

Thermodynamics underpins the energy conversion considered in this project. The model uses thermal efficiency to relate the thermal input to the electrical output, and the sensitivity analysis shows how this affects the resulting heat-rejection requirement.

7.2 Heat Transfer

Heat transfer occurs throughout the power-generation cycle, particularly during steam generation and condensation. Large heat-transfer duties are required to transfer energy between the heat source, working fluid and cooling system. The calculated waste-heat load of approximately 2,030 MW highlights the importance of effective heat rejection in a large thermal power plant.

7.3 Fluid Flow

Water and steam must be transported through the plant under controlled pressure and temperature conditions. Fluid-flow considerations therefore apply to piping, pumps, steam systems, cooling systems and associated equipment. Pressure losses and flow conditions would need to be considered in a detailed plant design.

7.4 Process Control

Although process control was not modelled directly in the Excel calculations, it would be important in a real plant for monitoring variables such as temperature, pressure and flow rate and maintaining them within operating limits..

7.5 Process Safety

Process safety is particularly important because the plant contains high-temperature and high-pressure systems alongside radioactive materials. A detailed safety assessment would consider hazards associated with pressure, temperature, loss of cooling, equipment failure and radioactive releases. These areas would require specialist safety analysis beyond the scope of this project.

7.6 Waste Management

Nuclear power generation produces radioactive materials that require controlled handling, treatment, storage and eventual disposal. Different waste streams can have different characteristics and activity levels, so their management would need to be considered separately in a detailed plant assessment.

7.7 Plant Design and Integration

A commercial power plant consists of many interconnected systems rather than a single piece of equipment. Equipment selection, piping, heat-transfer systems, pumps, controls and cooling systems must all be considered as part of the overall plant design. Detailed design

would require equipment sizing, mass and energy balances, hydraulic calculations and integration of the individual systems.

8. Nuclear Safety and Waste

Nuclear power plants require a high level of process and operational safety because they combine radioactive materials with high temperatures, high pressures and complex interconnected systems. Although detailed nuclear safety analysis is outside the scope of this project, several important principles were considered.

A key safety objective is to maintain control of the nuclear heat source and prevent the uncontrolled release of radioactive material. This requires multiple layers of protection, including physical barriers, monitoring systems, engineered safety systems and controlled operating procedures. [1] Maintaining adequate cooling is also important because heat continues to be generated following reactor shutdown.

From a process-engineering perspective, the steam and cooling systems are also significant. Equipment must be designed to operate safely under the expected pressure and temperature conditions, while instrumentation and control systems are required to monitor key process variables and identify abnormal operating conditions.

Radioactive waste is another important consideration. Different waste streams can have different levels of radioactivity and require different approaches to handling, treatment, storage and disposal. [2] Spent nuclear fuel and other higher-activity materials require particularly controlled management, while lower-activity operational wastes can be managed through different routes.

The project does not attempt to design a radioactive-waste facility or perform a detailed nuclear safety case. Instead, these considerations are identified as important areas that would require specialist analysis during the detailed design, operation and eventual decommissioning of a nuclear facility.

9. Limitations

The model was developed as a simplified process-engineering assessment rather than a detailed nuclear power plant design. Several assumptions were therefore made to keep the analysis manageable.

The thermal efficiency, capacity factor and emissions intensities were treated as fixed input values for the base case. In a real plant, performance would depend on operating conditions, equipment characteristics, plant design and maintenance requirements. The model also represents the overall energy conversion process using a single thermal efficiency rather than modelling individual components such as the steam generator, turbine, condenser and pump separately.

The waste-heat calculation is similarly simplified. It represents the difference between thermal input and electrical output and does not distinguish between different heat-rejection pathways or account for detailed cooling-system performance.

The emissions comparison is also indicative. The model applies assumed lifecycle emissions intensities to annual electricity generation rather than performing a full lifecycle assessment covering fuel production, construction, operation, decommissioning and waste management.

Finally, the project does not include detailed reactor physics, equipment sizing, heat-exchanger design, hydraulic calculations, nuclear safety analysis or radioactive-waste facility design. These areas would require significantly more detailed data and specialist engineering methods.

Despite these limitations, the model provides a useful first-level assessment of the relationships between thermal efficiency, electrical generation, waste heat and capacity factor, while demonstrating several chemical engineering principles relevant to nuclear power generation.

10. Conclusion

This project developed a simplified process-engineering assessment of a hypothetical 1 GW nuclear power plant using an Excel-based energy model. The model estimated the thermal power required to produce the target electrical output, the resulting waste-heat load and the annual electricity generation at an assumed capacity factor.

For the base case, a thermal efficiency of 33% resulted in an estimated thermal power requirement of approximately 3,030 MW and a waste-heat load of approximately 2,030 MW. At a capacity factor of 91%, the model estimated annual electricity generation of approximately 7.97 TWh.

Sensitivity analysis demonstrated that increasing thermal efficiency reduces the thermal power required and the associated waste-heat load, while increasing capacity factor increases annual electricity generation. An indicative emissions comparison was also performed using assumed lifecycle emissions intensities for nuclear and natural gas generation.

The project also considered the wider chemical engineering aspects of nuclear power, including thermodynamics, heat transfer, fluid flow, process control, process safety, waste management and plant integration. While the model is intentionally simplified and does not represent a detailed nuclear plant design, it demonstrates how fundamental chemical engineering principles can be applied to assess a complex energy-generation process.

Overall, the assessment applied basic energy calculations and sensitivity analysis to a nuclear power generation system while identifying the areas requiring more detailed specialist analysis in a real plant.

11: Nomenclature

- P_e : Electrical power output (MW)
- P_t : Required thermal power input (MW)
- η : Thermal efficiency (dimensionless / %)
- Q_{waste} : Waste-heat load to reject (MW)
- E : Annual electricity generation (TWh/year)
- H : Operating hours per year (h/year)
- CF : Plant capacity factor (dimensionless / %)
- CO_2e : Carbon dioxide equivalent

12. References

[1] Environment Agency (2026). *New nuclear power stations: assessing reactor designs*.

GOV.UK. Available at: <https://www.gov.uk/guidance/new-nuclear-power-stations-assessing-reactor-designs>

[2] Department for Energy Security and Net Zero (2025). *National Policy Statement for nuclear energy generation (EN-7)*. GOV.UK. Available at:

<https://www.gov.uk/government/publications/national-policy-statement-for-nuclear-energy-generation-en-7>